

# **ETC3550/ETC5550**

## **Applied forecasting**

Week 1: Introduction to forecasting & R



# Contact details

## Chief Examiner: Professor Rob Hyndman

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🏠 [robjhyndman.com](http://robjhyndman.com)

## Tutors

- **Mitchell O'Hara-Wild**
- Maliny Po
- Nuwani Palihawadana
- Xiefei (Sapphire) Li

# Brief bio

- Professor of Statistics, Monash University
- Co-author of most popular forecasting textbook in the world
- Lead developer of most popular forecasting software in the world

## How my forecasting methodology is used:

- Pharmaceutical Benefits Scheme
- Electricity demand
- Australian tourism demand
- Ageing population
- COVID-19 cases

# CASE STUDY 1: Paperware company

**Problem:** Want forecasts of each of hundreds of items. Series can be stationary, trended or seasonal. They currently have a large forecasting program written in-house but it doesn't seem to produce sensible forecasts. They want me to fix it.

## Additional information

- Program written in COBOL making numerical calculations limited. It is not possible to do any optimisation.
- Their programmer has little experience in numerical computing.
- They employ no statisticians and want the program to produce forecasts automatically.



# CASE STUDY 1: Paperware company

## Methods currently used

- A** 12 month average
- C** 6 month average
- E** straight line regression over last 12 months
- G** straight line regression over last 6 months
- H** average slope between last year's and this year's values. (Equivalent to differencing at lag 12 and taking mean.)
- I** Same as H except over 6 months.
- K** I couldn't understand the explanation.

## CASE STUDY 2: PBS



## CASE STUDY 2: PBS

**The Pharmaceutical Benefits Scheme (PBS) is the Australian government drugs subsidy scheme.**

- Many drugs bought from pharmacies are subsidised to allow more equitable access to modern drugs.
- The cost to government is determined by the number and types of drugs purchased. Currently nearly 1% of GDP.
- The total cost is budgeted based on forecasts of drug usage.

# CASE STUDY 2: PBS



# ABC News Online

AUSTRALIAN BROADCASTING CORPORATION



## NewsRadio

Streaming audio news

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Click "Refresh" or "Reload"  
on your browser for the latest edition.

This Bulletin: **Wed, May 30 2001 6:22 PM AEST**

## POLITICS

# Opp demands drug price restriction after PBS budget blow-out

The Federal Opposition has called for tighter controls on drug prices after the Pharmaceutical Benefits Scheme (PBS) budget blew out by almost \$800 million.

The money was spent on two new drugs including the controversial anti-smoking aid Zyban, which dropped

### the Public Record

For full election coverage

### FEATURES

### the Public Record

Federal Election 2001

For a fresh perspective on the federal election, reach into ABC Online's campaign weblog, [The Poll Vault](#).



## CASE STUDY 2: PBS

- In 2001: \$4.5 billion budget, under-forecasted by \$800 million.
- Thousands of products. Seasonal demand.
- Subject to covert marketing, volatile products, uncontrollable expenditure.
- Although monthly data available for 10 years, data are aggregated to annual values, and only the first three years are used in estimating the forecasts.
- All forecasts being done with the FORECAST function in MS-Excel!

## CASE STUDY 3: Car fleet company

**Client:** One of Australia's largest car fleet companies

**Problem:** how to forecast resale value of vehicles? How should this affect leasing and sales policies?

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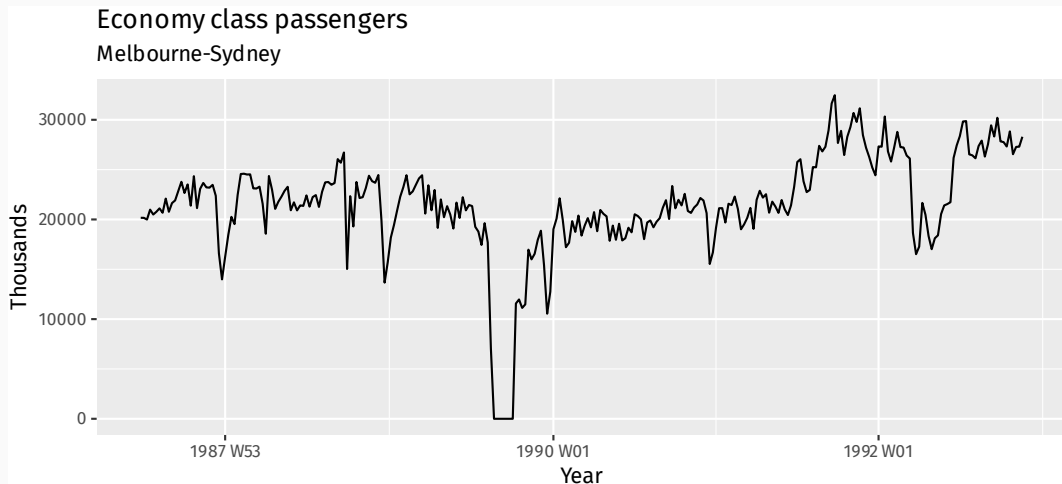
### Additional information

- They can provide a large amount of data on previous vehicles and their eventual resale values.
- The resale values are currently estimated by a group of specialists. They see me as a threat and do not cooperate.

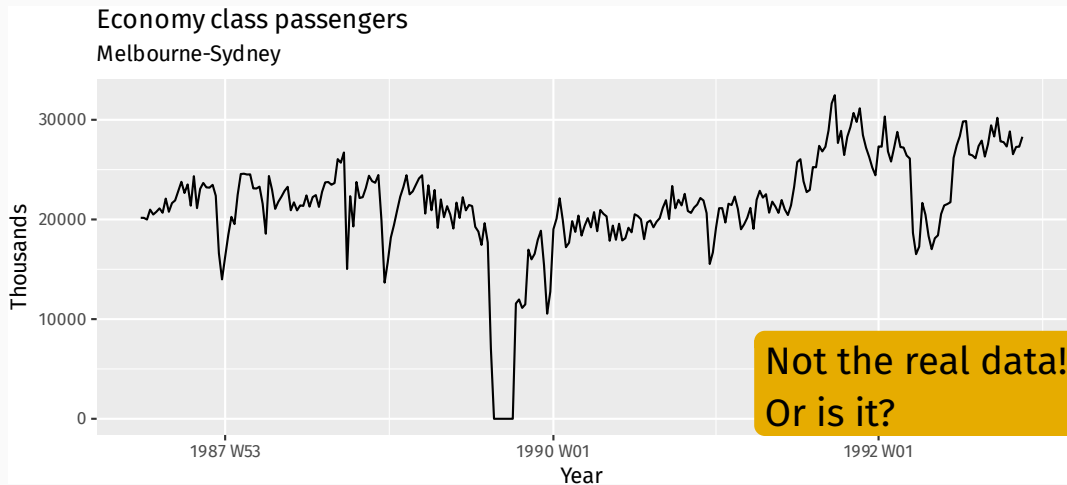
## CASE STUDY 4: Airline



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## CASE STUDY 4: Airline

**Problem:** how to forecast passenger traffic on major routes?

### Additional information

- They can provide a large amount of data on previous routes.
- Traffic is affected by school holidays, special events such as the Grand Prix, advertising campaigns, competition behaviour, etc.
- They have a highly capable team of people who are able to do most of the computing.

# Unit objectives

- 1 To obtain an understanding of common statistical methods used in business and economic forecasting.
- 2 To develop the computer skills required to forecast business and economic time series data;
- 3 To gain insights into the problems of implementing and operating large scale forecasting systems for use in business.



# Unit objectives

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## Teaching and learning approach

- Approximately one hour of pre-recorded online videos each week
- One hour online lecture each Monday
- One hour in-person workshop each Tuesday (focus on exam)
- One hour in-person tutorial each week (focus on assignments)

# Key reference

**Hyndman, R. J. & Athanasopoulos, G. (2021) *Forecasting: principles and practice*, 3rd edition**

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**[OTexts.com/fpp3/](https://otexts.com/fpp3/)**

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**[OTexts.com/fpp3/](https://otexts.com/fpp3/)**

- Free and online
- Data sets in associated R packages
- R code for examples
- Embedded online lectures

# Outline

| Week | Topic                               | Chapter |
|------|-------------------------------------|---------|
| 1    | Introduction to forecasting and R   | 1       |
| 2    | Time series graphics                | 2       |
| 3    | Time series decomposition           | 3       |
| 4    | Simple forecasting methods          | 5       |
| 5    | Accuracy evaluation                 | 5       |
| 6-7  | Exponential smoothing               | 8       |
| 8-10 | ARIMA models                        | 9       |
| 11   | Multiple regression and forecasting | 7       |
| 12   | Dynamic regression                  | 10      |

# Assessment

| Task                    | Due Date             | Value |
|-------------------------|----------------------|-------|
| Forecasting Competition | Fri 7 Mar            | 2%    |
| Weekly Quizzes          | end of weeks 2–11    | 8%    |
| Assignment 1            | Fri 28 Mar           | 6%    |
| Assignment 2            | Fri 25 Apr           | 6%    |
| Assignment 3            | Tue 20 May           | 6%    |
| Retail Project          | Fri 30 May           | 12%   |
| Final Exam              | Official exam period | 60%   |

# Assessment

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| Assignment 1            | Fri 28 Mar           | 6%    |
| Assignment 2            | Fri 25 Apr           | 6%    |
| Assignment 3            | Tue 20 May           | 6%    |
| Retail Project          | Fri 30 May           | 12%   |
| Final Exam              | Official exam period | 60%   |

- Need at least 45% for exam, and 50% for total.
- **ETC5550 students:** Extra project and exam questions.

**af.numbat.space**

- Includes all course materials
- Links for assignment submissions
- Link to discussion forum.

**Please don't send emails. Use the forum.**





- The IIF provides a prize to the top student in this subject each year.
- US\$100 plus one year membership.



Available for download from CRAN:

<https://cran.r-project.org>

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Available for download from RStudio:

<https://www.rstudio.com/products/rstudio/download/>

# Main packages



# Main packages

```
# Install required packages (do once)  
install.packages(c("tidyverse", "fpp3", "GGally"), dependencies = TRUE)
```

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install.packages(c("tidyverse", "fpp3", "GGally"), dependencies = TRUE)
```

```
# At the start of each session  
library(fpp3)
```

# Exercises Week 1

- Make sure you are familiar with R, RStudio and the tidyverse packages.
- Do the first five modules of `startr.numbat.space`.
- Forecasting competition

# Forecasting competition: forecast the following series

- 1 Google closing stock price on 24 March 2025
- 2 Maximum temperature at Melbourne airport on 14 April 2025
- 3 The difference in points (Collingwood minus Essendon) scored in the AFL match between Collingwood and Essendon for the Anzac Day clash. 25 April 2025
- 4 The seasonally adjusted estimate of total employment for April 2025. ABS CAT 6202, to be released around mid May 2025
- 5 Google closing stock price on 26 May 2025

**Due Friday 14 March**

For each of these, give a point forecast and an 80% prediction interval.

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## Due Friday 14 March

For each of these, give a point forecast and an 80% prediction interval.

Prize: \$50 cash prize



# Forecasting competition: scoring

$Y$  = actual,  $F$  = point forecast,  $[L, U]$  = prediction interval

## Point forecasts:

$$\text{Absolute Error} = |Y - F|$$

- Rank results for all students in class
- Add ranks across all five items

## Prediction intervals:

$$\text{Interval Score} = (U - L) + 10(L - Y)_+ + 10(Y - U)_+$$

- $u_+ = \max(u, 0)$
- Rank results for all students
- Add ranks across all five items

# tsibble objects

```
global_economy
```

```
# A tsibble: 15,150 x 6 [1Y]
```

```
# Key:           Country [263]
```

|    | Year  | Country     | GDP          | Imports | Exports | Population |
|----|-------|-------------|--------------|---------|---------|------------|
|    | <dbl> | <fct>       | <dbl>        | <dbl>   | <dbl>   | <dbl>      |
| 1  | 1960  | Afghanistan | 5377777811.  | 7.02    | 4.13    | 8996351    |
| 2  | 1961  | Afghanistan | 5488888896.  | 8.10    | 4.45    | 9166764    |
| 3  | 1962  | Afghanistan | 5466666678.  | 9.35    | 4.88    | 9345868    |
| 4  | 1963  | Afghanistan | 7511111191.  | 16.9    | 9.17    | 9533954    |
| 5  | 1964  | Afghanistan | 8000000044.  | 18.1    | 8.89    | 9731361    |
| 6  | 1965  | Afghanistan | 10066666638. | 21.4    | 11.3    | 9938414    |
| 7  | 1966  | Afghanistan | 13999999967. | 18.6    | 8.57    | 10152331   |
| 8  | 1967  | Afghanistan | 16733333418. | 14.2    | 6.77    | 10372630   |
| 9  | 1968  | Afghanistan | 13733333367. | 15.2    | 8.90    | 10604346   |
| 10 | 1969  | Afghanistan | 14088888922. | 15.0    | 10.1    | 10854428   |

```
# i 15,140 more rows
```

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|    | Year  | Country     | GDP                | Imports | Exports | Population |
|----|-------|-------------|--------------------|---------|---------|------------|
|    | Index | Key         | Measured variables |         |         |            |
| 1  | 1960  | Afghanistan | 537777811.         | 7.02    | 4.13    | 8996351    |
| 2  | 1961  | Afghanistan | 548888896.         | 8.10    | 4.45    | 9166764    |
| 3  | 1962  | Afghanistan | 546666678.         | 9.35    | 4.88    | 9345868    |
| 4  | 1963  | Afghanistan | 751111191.         | 16.9    | 9.17    | 9533954    |
| 5  | 1964  | Afghanistan | 800000044.         | 18.1    | 8.89    | 9731361    |
| 6  | 1965  | Afghanistan | 1006666638.        | 21.4    | 11.3    | 9938414    |
| 7  | 1966  | Afghanistan | 1399999967.        | 18.6    | 8.57    | 10152331   |
| 8  | 1967  | Afghanistan | 1673333418.        | 14.2    | 6.77    | 10372630   |
| 9  | 1968  | Afghanistan | 1373333367.        | 15.2    | 8.90    | 10604346   |
| 10 | 1969  | Afghanistan | 1408888922.        | 15.0    | 10.1    | 10854428   |

```
# i 15,140 more rows
```

# tsibble objects

```
tourism
```

```
# A tsibble: 24,320 x 5 [1Q]
# Key:           Region, State, Purpose [304]
  Quarter Region  State Purpose  Trips
   <qtr>  <chr>    <chr> <chr>    <dbl>
1 1998 Q1 Adelaide SA      Business 135.
2 1998 Q2 Adelaide SA      Business 110.
3 1998 Q3 Adelaide SA      Business 166.
4 1998 Q4 Adelaide SA      Business 127.
5 1999 Q1 Adelaide SA      Business 137.
6 1999 Q2 Adelaide SA      Business 200.
7 1999 Q3 Adelaide SA      Business 169.
8 1999 Q4 Adelaide SA      Business 134.
9 2000 Q1 Adelaide SA      Business 154.
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# i 24,310 more rows
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Domestic visitor  
nights in  
thousands by  
state/region and  
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```
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```
# A tsibble: 24,320 x 5 [1Q]  
# Key:      Region, State, Purpose [304]
```

|    | Quarter | Region   | State | Purpose  | Trips   |
|----|---------|----------|-------|----------|---------|
|    | Index   | Keys     |       |          | Measure |
| 1  | 1998 Q1 | Adelaide | SA    | Business | 135.    |
| 2  | 1998 Q2 | Adelaide | SA    | Business | 110.    |
| 3  | 1998 Q3 | Adelaide | SA    | Business | 166.    |
| 4  | 1998 Q4 | Adelaide | SA    | Business | 127.    |
| 5  | 1999 Q1 | Adelaide | SA    | Business | 137.    |
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# tsibble objects

- A `tsibble` allows storage and manipulation of multiple time series in R.
- It contains:
  - ▶ An index: time information about the observation
  - ▶ Measured variable(s): numbers of interest
  - ▶ Key variable(s): optional unique identifiers for each series
- It works with tidyverse functions.

# The tsibble index

Time index variables can be created with these functions:

| Frequency | Function |
|-----------|----------|
|-----------|----------|

|           |                               |
|-----------|-------------------------------|
| Annual    | <code>start:end</code>        |
| Quarterly | <code>yearquarter()</code>    |
| Monthly   | <code>yearmonth()</code>      |
| Weekly    | <code>yearweek()</code>       |
| Daily     | <code>as_date(), ymd()</code> |
| Sub-daily | <code>as_datetime()</code>    |